RESEARCH ARTICLE

Evidence before prediction: a configuration-driven framework for testing how molecular descriptors shape property models

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Abstract

Background. When a chemist builds a machine-learning model for a molecular property, most of the attention goes to the learner: a graph neural network, a descriptor-based model, or a hybrid. In practice much of the outcome is decided earlier, by the quality of the data, by which of several hundred descriptors enter the model, and by the role those descriptors are given. These choices are usually buried in task-specific scripts, so a reader cannot tell which were tested, which were assumed, and which produced the reported score.

Results. We present `chemistry-framework`, an open-source workflow in which an experiment is declared in one validated configuration file and every run writes a record that can be replayed. We tested three hypotheses. *H1, that training-partition evidence can guide descriptor choice*, is partly supported: subsets chosen from the scorecard beat random subsets of the same size at every budget tested, and significantly in 8 of 12 dataset–budget cells, but the rule ranked fourth of six selectors (mean rank 3.65). *H2, that the effect of adding descriptors transfers between implementations*, holds on four of five endpoints: the framework and Chemprop v2.3.1 agreed in direction and approximate size (+12.1% versus +19.7% on FreeSolv; at most about 2% elsewhere), while ESOL disagreed because one Chemprop run diverged and reached a test RMSE of 2.76 on a single split, a failure that only the per-split record exposed. *H3, that routes and predictive roles matter but are endpoint-specific*, holds: the full 210-descriptor hybrid led on ESOL and FreeSolv, the recommender route on Lipophilicity and graph-only on BACE, while on BBBP every route lay within 0.007 AUROC. No compact route separated from its matched random control.

Conclusions. Evidence from the training partition is useful for narrowing choices, but it does not name a universal winner, and a mean that hides one divergent run is unfinished evidence. Declaring the audit, the descriptor evidence, the route and the role in one configuration makes these comparisons cheap to run and possible to audit. The framework does not prescribe a better predictor; it makes the alternatives testable on the user's own data.

Keywords: Cheminformatics; molecular property prediction; feature selection; graph neural networks; QSAR; reproducibility; configuration-driven software

Background

A chemist who wants a model for a molecular property usually starts with a table: one row per compound, a SMILES string, and a measured value. Turning that table into a trustworthy prediction involves three decisions that are made before any model is fitted. The first is whether the data can carry the weight placed on them: are the structures valid, are they duplicated, do the labels agree, and does the training set cover the chemistry that will be predicted? The second is which of several hundred computed descriptors should enter the model, if any. The third is the role those descriptors should play: passed to the model as additional input, used as a fixed baseline, or used as the starting point that a graph network corrects.

In practice these three decisions are encoded in task-specific scripts. The script mixes the chemical judgement with the engineering plumbing, so a reader cannot tell which options were tested, which were assumed, and which

Table 1. Emphasis of related open-source molecular machine-learning tools, summarised from the cited publications. Entries are not exhaustive feature lists.

Tool	Primary emphasis	Feature handling	Roles of feature groups
Chemprop v2 [48]	Efficient, modular D-MPNN training and inference	Optional extra atom-, bond-, and molecule-level features	Direct (concatenated with the learned representation)
DeepMol [52]	Automated search over preprocessing, representation, and model pipelines	Feature extraction and selection as pipeline steps	Direct model input
QSARtuna [53]	Automated QSAR model and hyperparameter optimisation	Choice among descriptor and fingerprint representations	Direct model input
QSPRpred [55]	Reproducible, serialisable QSPR modelling, including proteochemometrics	Training-only variance, correlation, and Boruta filters	Direct model input
This work	Auditable link between feature evidence and model configuration	Training-partition evidence scorecard, dataset audit, relevance and redundancy recommender	Direct, fixed prior, or learned residual, declared per feature block

produced the reported number. Six months later the comparison is hard to repeat exactly. The problem is not a lack of algorithms. Descriptor-based models, fingerprints, and graph neural networks are all well developed, and benchmark collections such as MoleculeNet and the Therapeutics Data Commons allow systematic comparisons across endpoints [8]. Prior work shows that descriptor-based models can match or outperform graph models on some tasks, and that adding descriptors to a learned graph representation can help on others. The recurring lesson is that the usefulness of a representation depends on the endpoint, the curation of the data, and the evaluation protocol, not on the model class alone.

Decisions about features are equally consequential. Quantitative structure–activity and structure–property relationship (QSAR/QSPR) guidance asks for a defined endpoint, a transparent algorithm, and a stated applicability domain. Feature selectors address part of this: filters rank descriptors by correlation or mutual information, while embedded methods use a fitted learner, such as Random Forest importance or sparse regression, to rank them. Every such ranking is evidence for a particular dataset, learner, and budget. It is not a universal statement about which molecular properties matter, and presenting it as one is a common source of over-claiming.

A related decision is how the selected information enters the predictor. Descriptors may be concatenated with a learned graph representation, used as a fixed linear prior learned on the training set, or used as the baseline that the graph model corrects. These roles answer different modelling questions, yet they are usually implemented in separate scripts or frozen into a pipeline, which makes an otherwise simple comparison expensive.

Open-source software already supplies much of the infrastructure. Chemprop provides efficient directed message-passing networks [48], DeepMol and QSARtuna automate pipeline and model search [52, 53], QSPRpred emphasises reproducible and serialisable QSPR workflows [55], and ZairaChem targets automated screening cascades. General tools such as Hydra and MLflow record settings and runs. The gap addressed here is narrower and, we argue, more useful to a chemist: none of these tools makes training-partition feature evidence, descriptor recommendations, feature routes, and predictive roles explicit inside one auditable experiment record (Table 1).

We therefore built `chemistry-framework`, an open-source Python package in which an experiment is declared in one plain-text configuration file. The file names the dataset, the molecular representations, the split, the model, and the role of each descriptor block; it is checked against a typed schema before any computation starts. Fig. 1 shows the five stages a user goes through, from a table of SMILES and measured values to a saved record that another person can replay. The framework does not replace the chemist's judgement. It makes each judgement visible, testable, and cheap to change.

What this study tests

The paper is organised around three hypotheses, stated here so that the reader can judge the results against them rather than after the fact.

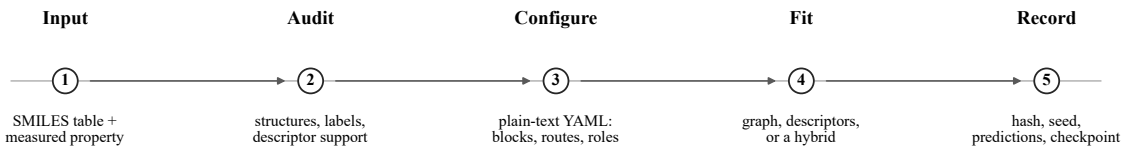


Fig. 1. The five stages of a `chemistry-framework` experiment. A table of SMILES and measured properties is read and audited, the experiment is declared in a validated configuration file, the selected representations are fitted, and every run writes a record (configuration hash, split seed, predictions, checkpoint) that a reader can inspect and re-run. The framework changes what is explicit, not what is chemically possible.

Table 2. Reading map: each hypothesis, the experiment that tests it, and where the answer is reported.

	Hypothesis	Experiment	Reported in
H1	Training-partition evidence guides descriptor choice	Scorecard and dataset audit on five endpoints; six selectors at four budgets with a Random Forest downstream; recommended versus random subsets	Descriptor evidence and data audit; Tables 5 and 6; Fig. 3
H2	The descriptor effect transfers between implementations	Graph-only and graph-plus-descriptor models in `chemistry-framework` and in Chemprop v2.3.1, on the same saved splits	Predictive performance; Tables 3 and 4; Fig. 2
H3	Route and role matter, but the best choice is endpoint-specific	Five role configurations; a harmonised routing experiment with Spearman, mutual-information and random controls; per-split paired comparisons	Descriptor routing and predictive roles; Tables 7 and 8; Fig. 4

H1 — Evidence is usable. Training-partition diagnostics of descriptors, together with a transparent relevance-and-redundancy rule, carry enough information to guide descriptor choice. *Prediction:* descriptor subsets chosen from this evidence should beat random subsets of the same size more often than not, and the choice should be stable across splits. *Falsified if:* the evidence-based choice is indistinguishable from a random one.

H2 — The descriptor effect transfers. The effect of adding descriptors to a graph model is a property of the endpoint and the data, not of one software implementation. *Prediction:* an independent implementation, Chemprop, should change in the same direction and by a similar amount on most endpoints. *Falsified if:* the two implementations disagree in sign across endpoints.

H3 — Route and role matter, but not universally. Compact descriptor routes and alternative predictive roles change results, yet the best choice is endpoint-specific and small compared with split-to-split variation. *Prediction:* no single route or role wins everywhere, and most differences should be comparable to the between-split standard deviation. *Falsified if:* one configuration wins on every endpoint by a margin that exceeds the split variation.

Testing these hypotheses requires more than a new script. It requires the same splits, the same checkpoints, and the same evaluation rule across every model, and it requires that the comparison be reproducible by someone who did not run it. The remainder of the paper reports the design (Methods), the results for each hypothesis (Results), what they do and do not imply for a practising chemist (Discussion), and the conclusions (Conclusions). Table 2 maps each hypothesis to its experiment and to the section and table where the evidence is reported.

Results

Performance depends on the endpoint, not on the model class

The best model depended on the task (Tables 3 and 4). The descriptor MLP and the framework direct hybrid tied for the lowest mean RMSE on ESOL (0.623), the framework direct hybrid was best on Lipophilicity (0.606), and Chemprop's

Table 3. Regression performance (RMSE, mean ± SD over ten saved splits). Bold marks the best mean in each column; ties at the displayed precision are both bold.

Model	ESOL	FreeSolv	Lipophilicity
Reference models			
Dummy	1.894 ± 0.197	3.619 ± 0.388	1.148 ± 0.039
ECFP4 + Random Forest	1.352 ± 0.093	1.795 ± 0.306	0.853 ± 0.028
RDKit 2D + Random Forest	0.637 ± 0.061	1.089 ± 0.175	0.748 ± 0.025
Descriptor MLP	$\mathbf{0.623 \pm 0.040}$	1.290 ± 0.446	0.683 ± 0.023
Chemprop v2.3.1			
Graph-only D-MPNN	0.791 ± 0.063	1.178 ± 0.320	0.641 ± 0.085
D-MPNN + descriptors (x_d)	0.844 ± 0.640	$\mathbf{0.946 \pm 0.261}$	0.639 ± 0.068
Framework			
Graph only (`residual_only`)	0.623 ± 0.092	1.251 ± 0.153	0.607 ± 0.031
Graph + descriptors (`direct_hybrid`)	$\mathbf{0.623 \pm 0.054}$	1.099 ± 0.249	$\mathbf{0.606 \pm 0.019}$
Prior + graph residual (`prior_residual`)	0.681 ± 0.113	1.175 ± 0.207	0.609 ± 0.036

Table 4. Classification performance (AUROC, mean ± SD over ten saved splits). Bold marks the best mean in each column; ties at the displayed precision are both bold.

Model	BBBP	BACE
Reference models		
Dummy	0.500 ± 0.000	0.500 ± 0.000
ECFP4 + Random Forest	0.905 ± 0.022	0.822 ± 0.028
RDKit 2D + Random Forest	0.912 ± 0.019	$\mathbf{0.822 \pm 0.027}$
Descriptor MLP	0.915 ± 0.028	0.787 ± 0.034
Chemprop v2.3.1		
Graph-only D-MPNN	0.911 ± 0.024	0.808 ± 0.056
D-MPNN + descriptors (x_d)	$\mathbf{0.923 \pm 0.019}$	0.793 ± 0.040
Framework		
Graph only (`residual_only`)	0.914 ± 0.034	0.802 ± 0.038
Graph + descriptors (`direct_hybrid`)	0.916 ± 0.029	0.785 ± 0.033
Prior + graph residual (`prior_residual`)	0.910 ± 0.034	0.783 ± 0.062

D-MPNN with descriptors was best on FreeSolv (0.946) and BBBP (0.923); the RDKit 2D Random Forest remained best on BACE (0.822). After Holm correction only two paired comparisons excluded zero: the framework hybrid beat the RDKit 2D Random Forest on Lipophilicity ($\Delta = -0.142$, corrected 95% CI $[-0.179, -0.106]$, 10/10 splits) and trailed it on BACE (hybrid -0.037 AUROC, $[-0.058, -0.017]$, 0/10 wins); every other comparison was inconclusive, including the ESOL hybrid-versus-Random-Forest and hybrid-versus-graph-only contrasts (Additional file 1, Table S5). Descriptors outperformed ECFP4 fingerprints with Random Forests on every task, although the BACE margin was negligible (0.822 versus 0.822 at the displayed precision).

The framework and Chemprop graph only models were close: the framework was more accurate on ESOL (0.623 versus 0.791, nominal $p = 0.027$, Holm-adjusted $p = 0.053$ within the two external comparisons for that dataset), Chemprop was more accurate on FreeSolv (1.252 versus 1.178, not significant), and the remaining tasks were indistinguishable; across tasks the framework won between three and nine of ten splits (Additional file 1, Table S6). More informative than either ranking was the effect of adding descriptors (Fig. 2). In the framework the change in the mean was small on ESOL (+0.1%), Lipophilicity (+0.1%) and BBBP (+0.2%), clearly positive on FreeSolv (+12.1%), and slightly negative on BACE (−2.1%). Chemprop showed the same pattern on FreeSolv (+19.7%), Lipophilicity (+0.2%), BBBP (+1.3%) and BACE (−1.9%), but on ESOL its descriptor hybrid was better on nine of the ten splits and failed badly on one (test RMSE 2.76), so the mean fell by 6.8% with an SD of 0.64. Reporting only that mean would have reversed the sign of the effect on ESOL; the per-split record kept by the workflow is what makes the failure visible. On the four endpoints without such a failure the two implementations agreed on the sign and approximate size of the descriptor effect, which suggests that the effect is a property of the task and protocol rather than of the code.

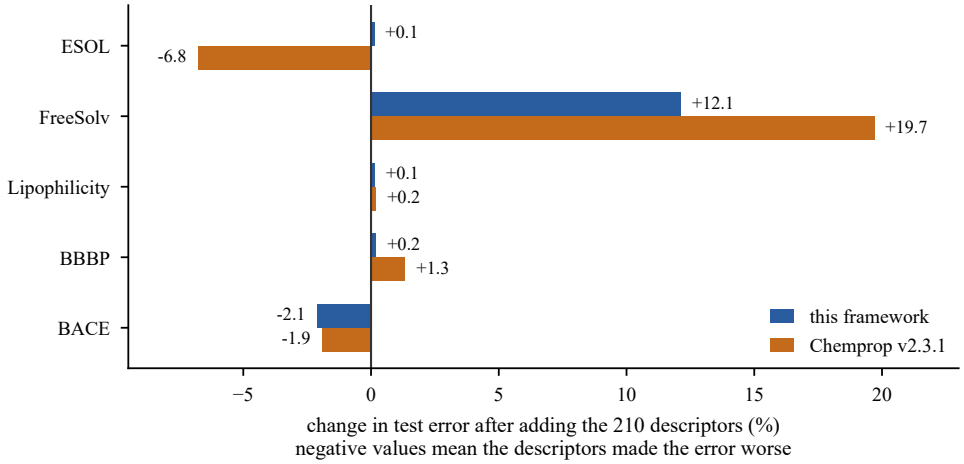


Fig. 2. Effect of adding 210 RDKit descriptors to a graph-only D-MPNN, in two independent implementations. Bars show the change in the ten-split mean of the primary metric (RMSE for ESOL, FreeSolv and Lipophilicity; AUROC for BBBP and BACE), signed so that positive values mean a lower error or a higher AUROC. The ESOL value for Chemprop is dominated by one split, in which its descriptor hybrid reached a test RMSE of 2.76 while being better than graph-only on the other nine. Computed from Tables 3 and 4.

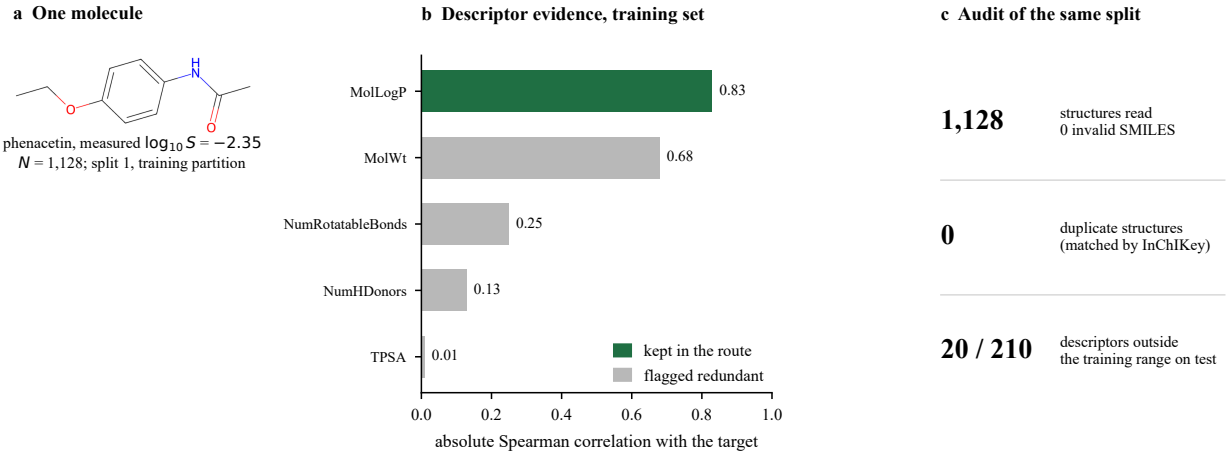


Fig. 3. One molecule followed through the audit and the descriptor scorecard. (a) Phenacetin in the ESOL dataset; its measured aqueous solubility is $\log_{10} S = -2.35$ and it belongs to the training partition of split 1. (b) Training-partition evidence for five of the 210 descriptors. Bars are absolute Spearman correlations with the measured solubility; a descriptor is grey if the recommender did not keep it, which in these cases is because its Pearson correlation with a higher-ranked descriptor exceeded 0.85. (c) Audit output for the same split. The full scorecard covers every descriptor and every split.

Descriptor evidence shows what the data support, not what is universally important

A chemist approaching a new endpoint asks two things before any model is trained: are the records trustworthy, and which of the several hundred descriptors are actually supported by the training molecules? The audit answers the first, the scorecard and the selectors answer the second, and the two answers are independent of the learner. The audit found duplicated structures in two of the five benchmarks (Table 6, discussed below).

Figure 3 follows a single molecule through both checks, so that the numbers in the tables can be read on a concrete case.

With a Random Forest as learner (ESOL, FreeSolv, BBBP), recommended subsets had better mean scores than random subsets of the same size at every budget, and the corrected intervals excluded zero at all budgets on ESOL, at $K \leq 20$ on FreeSolv, and at $K = 5$ on BBBP; the margin shrank as K grew. At $K = 50$ on ESOL, 10 and 20 on FreeSolv, and the three larger budgets on BBBP, the recommendation still had the better mean but the corrected interval did not separate it from random draws. Against the full 210-descriptor pool the picture reverses: at $K = 5$

Table 5. Six feature selectors on five datasets with a Random Forest downstream. Mean rank (1 = best) per dataset over $K \in \{5, 10, 20, 50\}$ and overall over 20 dataset-budget cells; Best denotes cells in which the selector had the best mean score; Jaccard: mean pairwise stability over splits and cells. Full results in Additional file 1, Table S4.

	Mean rank per dataset							
Selector	ESOL	FreeSolv	Lipo.	BBBP	BACE	Overall	Best	Jaccard
RF impurity importance	1.0	1.8	1.8	4.8	1.8	2.20	10	0.65
Mutual information	2.8	2.5	1.5	2.2	2.8	2.35	3	0.80
MI-Pearson	3.8	4.2	3.5	1.5	2.8	3.15	4	0.57
Framework recommender	3.8	3.2	3.5	3.8	4.0	3.65	1	0.76
Spearman top-K	3.8	3.2	5.0	5.8	5.8	4.70	0	0.88
L1 (lasso / L1-logistic)	6.0	6.0	5.8	3.0	4.0	4.95	2	0.47

Table 6. Structure-level audit by InChIKey. Duplicated structures are structures recorded more than once (records in parentheses). Conflicts are duplicated structures whose labels disagree (classification) or span more than half the target standard deviation (regression).

Dataset	Records	Unique InChIKeys	Duplicated structures	Conflicts	Largest span
ESOL	1,128	1,117	11 (22)	0	1.03
FreeSolv	642	642	0 (0)	0	0.00
Lipophilicity	4,200	4,200	0 (0)	0	0.00
BBBP	2,039	1,971	64 (132)	11	–
BACE	1,513	1,513	0 (0)	0	–

and $K = 10$ on ESOL the recommended subset was significantly worse, and every other budget was inconclusive (Additional file 1, Table S3).

Across six selectors on five datasets (Table 5), Random Forest importance and mutual information ranked best on average, with mean ranks of 2.20 and 2.35, respectively. The framework recommender ranked fourth (3.65), was best in one of the 20 dataset-budget cells, and had mean pairwise Jaccard stability of 0.76. This stability was below Spearman ranking (0.88) and mutual information (0.80), but above Random Forest importance (0.65), MI–Pearson (0.57), and L1 (0.47). These results support a transparent and reasonably stable candidate-generation heuristic, not a selector that consistently outperforms established alternatives. The selector match with the downstream Random Forest may favour RF importance.

The practical reading for a chemist is therefore: the scorecard tells you what the data can support, the recommender proposes a compact and stable subset, and the ranking among selectors is close enough that the choice should be made on inspectability and stability rather than on a small score difference.

The data audit finds problems that the training score cannot see

The audit found duplicated structures in two of the five benchmarks (Table 6). BBBP contained 64 structures recorded more than once (132 records, 6.5%), 11 of them with conflicting labels; ESOL contained 11 duplicated structures whose label spans stayed below the conflict threshold (largest span 1.03 log units, compared with a threshold of 1.0478). Across the ten scaffold splits, the mean number of out-of-range descriptors was 19.3 ± 12.7 for ESOL, 20.8 ± 14.8 for Lipophilicity, and 15.0 ± 9.6 for BBBP; the corresponding fractions of test molecules affected were $6.3 \pm 2.5\%$, $3.7 \pm 0.7\%$, and $3.7 \pm 2.1\%$, respectively. These diagnostics are range-based indicators rather than a comprehensive measure of distribution shift. In BBBP the relationship was not monotone: the hybrid's weakest split (split 8) had 19 out-of-range descriptors and 8.8% of test molecules affected, but split 3 had almost as many (18) and gave the hybrid its best score (Additional file 1, Table S2). A range-based count is therefore a warning flag, not a predictor of which split will be hardest.

The duplicate-controlled BBBP sensitivity analysis retained 1,960 unique structures after excluding the 11 conflicting label groups and retaining one record for each of the 53 remaining duplicated groups. On ten new scaffold manifests, mean AUROC was 0.8656 ± 0.0577 for the framework graph only model and 0.8637 ± 0.0422 for its descriptor hybrid. Chemprop v2.3.1 achieved 0.8686 ± 0.0398 with graph only and 0.8744 ± 0.0428 with the 210 molecular

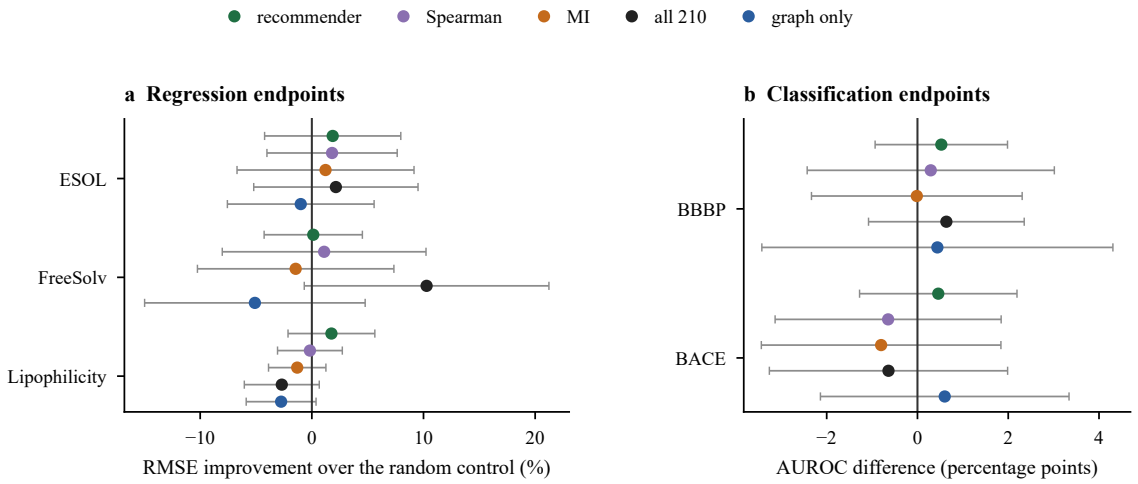


Fig. 4. Compact descriptor routes compared with the matched random control. Points are mean paired differences over ten splits and error bars are population SDs. Positive values mean that the route had a lower RMSE (a) or a higher AUROC (b) than a random subset of the same size drawn at the same split-specific budget K^*. The largest mean improvement is the full descriptor set on FreeSolv (about 10%), but every interval includes zero: at this sample size no route is separated from its control.

descriptors, while RDKit 2D plus Random Forest achieved 0.8940 ± 0.0305 (Additional file 1, Table S12). The hybrid minus graph mean difference was -0.0019 for the framework (5/10 splits favoured the hybrid) and $+0.0057$ for Chemprop (9/10 splits favoured the hybrid). For the framework the paired 95% t interval included zero, so these results do not establish a conclusive advantage from descriptor augmentation in this workflow. The Chemprop hybrid-minus-graph difference was in the same direction but smaller, and its interval also included zero; after Holm correction within the five sensitivity comparisons neither implementation separates (Additional file 1, Table S12b). The exact ranking differed from the primary benchmark. Scores were lower for the learned models on the sensitivity data, but this change cannot be attributed to duplicate leakage alone because both data composition and split manifests differed.

Descriptor routing depends on feature budget and endpoint

The harmonised routing results show that the value of a compact descriptor route depends on the endpoint (Table 7). The full 210-descriptor hybrid had the lowest mean RMSE on ESOL (0.614 ± 0.039) and FreeSolv (1.038 ± 0.214). On Lipophilicity the recommender route had the lowest mean of the six routes (0.581 ± 0.041), below the full descriptor set (0.606); on BACE the graph-only model had the highest mean AUROC (0.798), and the six routes spanned only 0.014, less than the between-split SD. On BBBP every route lay within 0.007 AUROC of the best. The random-subset control is the important comparator. The recommender beat it on Lipophilicity (0.581 versus 0.591), but was within one SD of it on ESOL (0.620 versus 0.631), FreeSolv (1.147 versus 1.149), BBBP (0.915 versus 0.910) and BACE (0.797 versus 0.792), and every paired recommender-minus-control difference had a Holm-corrected interval that included zero (Additional file 1, Table S7b). The same correction left the Lipophilicity gap to the full descriptor set inconclusive, so the apparent benefit of compactness is not established at this sample size either. A compact route therefore does not appear to require an informed selection rule on these endpoints; what matters is that it is tested rather than assumed. Per-split outputs and checkpoint decisions are archived in the repository; the checkpoint selection rule and result-file inventory are documented in Additional file 1, Table S13.

Configurable predictive roles are useful but not uniformly beneficial

The best role assignment was also task dependent (Table 8). The descriptor-only role was best on ESOL (0.619), direct fusion on FreeSolv, Lipophilicity, and BBBP, and graph only on BACE (0.802 versus 0.785 for direct fusion). The margins were often small: on Lipophilicity and ESOL the four learned roles other than the prior and prior-plus-

Table 7. Harmonised evidence-guided descriptor routing. Entries are test RMSE for ESOL, FreeSolv, and Lipophilicity, and test AUROC for BBBP and BACE; values are mean ± population SD over ten saved splits. All route checkpoints are selected by minimum validation loss, and the recommended route's K^* is selected using the primary validation metric at that checkpoint. Spearman and MI use the same split-specific K^*; the random route averages ten random-subset draws per split. Bold text marks the best mean in each row. The full checkpoint-level audit covers 250 model runs; see Additional file 1, Table S13.

Dataset	Metric	Graph only	All 210	Recommender	Spearman	MI	Random
ESOL	RMSE ↓	0.638 ± 0.074	**0.614 ± 0.039**	0.620 ± 0.078	0.620 ± 0.075	0.622 ± 0.062	0.631 ± 0.057
FreeSolv	RMSE ↓	1.205 ± 0.164	**1.038 ± 0.214**	1.147 ± 0.140	1.134 ± 0.156	1.168 ± 0.180	1.149 ± 0.141
Lipophilicity	RMSE ↓	0.607 ± 0.031	0.606 ± 0.019	**0.581 ± 0.041**	0.592 ± 0.033	0.599 ± 0.032	0.591 ± 0.027
BBBP	AUROC ↑	0.914 ± 0.034	**0.916 ± 0.029**	0.915 ± 0.023	0.913 ± 0.020	0.910 ± 0.018	0.910 ± 0.020
BACE	AUROC ↑	**0.798 ± 0.038**	0.786 ± 0.033	0.797 ± 0.036	0.786 ± 0.059	0.784 ± 0.051	0.792 ± 0.042

Table 8. Predictive-role ablation (primary metric, mean ± SD over ten splits; RMSE for ESOL, FreeSolv, and Lipophilicity, AUROC for BBBP and BACE). Bold text marks the best mean per column.

Configuration	ESOL	FreeSolv	Lipophilicity	BBBP	BACE
`prior_only`	0.913 ± 0.073	2.015 ± 0.116	1.009 ± 0.036	0.812 ± 0.030	0.660 ± 0.055
`direct_only`	**0.619 ± 0.045**	1.350 ± 0.458	0.683 ± 0.034	0.903 ± 0.041	0.785 ± 0.046
`residual_only`	0.623 ± 0.092	1.251 ± 0.153	0.607 ± 0.031	0.914 ± 0.034	**0.802 ± 0.038**
`direct_hybrid`	0.623 ± 0.054	**1.099 ± 0.249**	**0.606 ± 0.019**	**0.916 ± 0.029**	0.785 ± 0.033
`prior_residual`	0.681 ± 0.113	1.175 ± 0.207	0.609 ± 0.036	0.910 ± 0.034	0.783 ± 0.062

residual variants spanned 0.004 and 0.018 RMSE respectively, so the ordering should be read as a trend across ten splits rather than as a separated ranking. No paired interval excluded zero for the prior/residual model against graph only on any task, and on ESOL it had the largest split-to-split variability of the learned models. At the level of training seeds, variation was small relative to split variation for most models (SD over three seeds 0.003 - 0.007 AUROC on BBBP), but reached 0.013 for the prior/residual model, about a third of its between-split SD of 0.034 (Additional file 1, Table S8).

Discussion

The results support an evidence-guided approach to molecular property modelling, rather than a universal ranking of model families. Across five independent endpoints, descriptor-based, graph-based, and hybrid predictors each had tasks on which they performed well. Adding the 210 RDKit descriptors changed performance in the same direction in the framework and Chemprop implementations on four of the five tasks, and by a similar amount: a clear gain on FreeSolv and a change of a few percent or less elsewhere. On ESOL the two disagreed, but only because one Chemprop run diverged: its hybrid was better on nine of ten splits. This cross-implementation agreement, and the single visible failure, support treating descriptor utility as an empirical property of a task and protocol, and as a reason to keep per-split records rather than only summaries.

It is worth stating what this means for the chemist who has one dataset in hand. The default "use the graph" or "use all descriptors" habit is not wrong, but it is a guess, and the cost of testing it is now small. On phenacetin's endpoint the same graph encoder gave a test RMSE of 0.681 without descriptors and 0.595 with them on that split, a difference of 0.086; averaged over ten splits the graph-only and full-descriptor means are within 0.001 RMSE of each other on ESOL (0.623 versus 0.623), which is exactly why a single split should not be used as a decision rule. The same holds for compact routes: the recommender gave the lowest mean on Lipophilicity (0.581 versus 0.591 for random subsets), although the paired difference did not survive correction, and it did not separate from the control on any other endpoint. None of these numbers is a universal rule, and the framework deliberately does not turn any of them into one. What it does is make the alternatives cheap enough to compare on the user's own data, with the same split and the same record.

The first contribution is the coupling of feature evidence to model design, with a deliberate distinction between documentation and recommendation. The scorecard is an evidence-documentation and diagnostic layer: it records feature provenance, training support, target association, redundancy, and post-hoc range coverage, while the dataset

audit makes duplicate structures and conflicting labels visible. The recommender is a separate candidate-generation heuristic that applies a fixed relevance and redundancy rule; it does not consume categorical scorecard evidence or automatically determine model roles. The recommender was reasonably stable across splits and often improved on random descriptor subsets when a Random Forest was trained directly on the selected descriptors. However, it did not outperform all selectors: Random Forest importance and mutual information ranked higher on average, and the recommender was not reliably superior to random subsets inside graph hybrids. We therefore interpret it as a transparent decision aid, not a new state-of-the-art selector or a measure of universal chemical importance. The scorecard's audit value does not depend on the recommender being the best-performing selector.

The second contribution is a configuration-driven modular architecture. Named feature blocks can be separated from routes and assigned to direct, fixed-prior, or learned-residual roles without changing feature definitions. The role ablation showed why this separation matters: direct fusion had the best mean on three of five tasks and the descriptor-only role on one, while graph only was best on BACE and within 0.001 RMSE of the best route on Lipophilicity, so the ordering is endpoint-specific and often within between-split noise. The prior and residual design did not produce a statistically conclusive gain over graph only. Its value in this study is therefore architectural flexibility and a controlled way to test role assignments, rather than a claim that this role is consistently more accurate.

The third contribution is evaluation across molecular representations and different prediction endpoints. The same workflow accommodates fingerprints, computed descriptors, graph encoders, and hybrid inputs, and the benchmark spans scalar regression and binary classification. On the duplicate-controlled blood-brain barrier dataset, the framework and Chemprop hybrids had similar mean area under the receiver operating characteristic curve, while the Random Forest on RDKit descriptors had the highest mean. The exact ranking differed from the primary benchmark. Because the cleaned dataset and its split manifests also differed, this sensitivity analysis does not isolate duplicate leakage as the sole cause of the score change. Likewise, the benchmarks are separately trained single-endpoint experiments; they demonstrate cross-task workflow coverage, not the benefit of joint multi-task learning.

Several limitations remain. Ten split manifests still provide limited statistical power for differences of a few hundredths of a metric unit, and overlapping training sets make paired intervals descriptive rather than definitive. The primary blood-brain barrier benchmark retained duplicate structures, including groups with conflicting labels, and the audit found that tautomeric representations can produce different scaffold strings for the same InChIKey. The duplicate-controlled analysis addresses sensitivity to curation, but changes to data composition and splits prevent a causal attribution to leakage alone. Hyperparameters were fixed rather than exhaustively tuned. The descriptor-only reference and the framework's direct descriptor model were produced by different runners. The routing experiment now uses a harmonised checkpoint-selection rule across all five endpoints, with the complete checkpoint-level audit available in the repository. The ten-split summaries are descriptive and should not be interpreted as definitive evidence of small performance differences. Finally, the experiments use two-dimensional inputs and cover only regression and binary classification. Future work should evaluate additional datasets and prospective splits, strengthen duplicate-aware partitioning, test joint multi-task training where supported, and assess whether alternative feature evidence improves decisions beyond the present transparent heuristic.

Conclusions

We stated three hypotheses in the Background so that they could fail. The results answer them as follows.

H1, evidence is usable: partly supported. Training-partition evidence is informative enough to act on. With a Random Forest downstream, descriptor subsets chosen from the scorecard beat random subsets of the same size at every budget tested, and the difference survived correction in 8 of the 12 dataset–budget cells. It is not, however, decisive: the recommender ranked fourth of six selectors overall (mean rank 3.65), and the correlation with the target is a weak guide to which descriptors a model will actually use. The honest description is a stable and inspectable way to generate a compact candidate set, not a claim that the selected descriptors are the important ones.

H2, the descriptor effect transfers: supported on four of five endpoints. Adding 210 descriptors to a graph model changed the error by a similar amount in two independently written implementations on FreeSolv (+12.1% and +19.7%), Lipophilicity (+0.1% and +0.2%), BBBP (+0.2% and +1.3%) and BACE (−2.1% and −1.9%). ESOL was the exception, and it is instructive: Chemprop's descriptor hybrid was better than graph-only on nine of ten splits and failed on one (test RMSE 2.76), so the ten-split mean reversed the sign of the effect. A mean without its per-split record would have produced a confident and wrong statement about chemistry. The practical consequence is that a descriptor effect should be reported per split before it is interpreted.

H3, route and role matter but are endpoint-specific: supported. The best route differed by endpoint: the full 210-descriptor hybrid on ESOL and FreeSolv, the recommender on Lipophilicity, graph-only on BACE, and no distinguishable winner on BBBP. The best predictive role also changed with the endpoint. Crucially, no compact route separated from a random subset of the same size once paired uncertainty was included, so the route is worth testing but should not be sold as a source of large gains on these endpoints.

Taken together, the three answers point to a practical rule for a chemist: test the alternatives on your own splits, keep the per-split record, and treat a mean that hides one divergent run as unfinished. `chemistry-framework` exists to make that rule cheap to follow. An experiment is a plain-text configuration file; the five commands behind it read the data, audit it, validate the configuration, fit the declared representations, and write a record containing the configuration hash, split seed, predictions and checkpoint. A colleague or referee can therefore see which choice was made, on which evidence, and what happened when it was changed. The framework does not promise a universally better predictor, and the results here give no reason to expect one.

Supplementary information

Additional file 1: Supplementary results (PDF). Dataset overview, target statistics, and provenance (Table S1a); fixed hyperparameters (Table S1b); scorecard diagnostics and split-level shift analysis (Table S2); recommended versus random subsets and the full pool (Table S3); six-selector results, stability and budget curves (Tables S4 and S4b, Figure S1); internal and external paired comparisons (Tables S5–S6); harmonised descriptor-routing results and their paired comparisons (Tables S7 and S7b); training-seed variance (Table S8); secondary metrics and per-split scores (Tables S9–S10); advisor case study on Caco-2 (Table S11); duplicate-controlled BBBP sensitivity analysis and its paired comparisons (Tables S12 and S12b); checkpoint protocol and result-file inventory (Table S13).

Abbreviations

AP, average precision; AUROC, area under the receiver operating characteristic curve; BACE-1, beta-secretase 1; BBBP, blood–brain barrier penetration; CI, confidence interval; D-MPNN, directed message-passing neural network; ECFP, extended-connectivity fingerprint; GIN, graph isomorphism network; GNN, graph neural network; InChI, International Chemical Identifier; InChIKey, International Chemical Identifier key; LARS, least angle regression; MAE, mean absolute error; MI, mutual information; MI-Pearson, selector using mutual-information relevance and Pearson-correlation redundancy; MLP, multilayer perceptron; QSPR, quantitative structure–property relationship; QSAR, quantitative structure–activity relationship; RF, Random Forest; RMSE, root mean squared error; SD, standard deviation; SMILES, simplified molecular input line entry system; TDC, Therapeutics Data Commons.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The datasets are publicly available through MoleculeNet [8]. Split manifests, configuration files, per-split results, the scripts that produced them, and the manuscript sources are available at `https://github.com/soratori1025/Framework-Chemistry` (branch `paper1`, archived for this submission). Documentation, including a quick-start guide and the configuration reference, is maintained in the same repository.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

D.K.N.N. conceived the framework, developed the software, performed the experiments, analysed the results, and wrote the manuscript.

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Methods

Framework and configuration

`chemistry-framework` organises an experiment as data loading, configuration validation, splitting, featurisation, feature analysis, model construction, training, and evaluation (Fig. 1). Molecules are supplied as strings in the simplified molecular input line entry system (SMILES). A YAML (human-readable data serialisation) configuration declares the input file and targets, named feature blocks, routes that assign blocks to tasks, priors, task types, split strategy, and model, training, evaluation, and analysis settings; it is parsed against a typed schema before any computation. Feature definitions and resolved feature names are hashed, so cached matrices and results can be traced to the definitions that produced them. Feature blocks include element counts, bond-pair counts, substructure counts from SMILES arbitrary target specification (SMARTS) patterns, Morgan fingerprints [3], and the RDKit 2D descriptor set (210 descriptors in RDKit 2023.09.6) [60]; model components, implemented in PyTorch [61], include directed message-passing neural network (D-MPNN) and graph isomorphism network (GIN) encoders, a multilayer perceptron (MLP) descriptor encoder, fusion modules, and task-specific heads. A command-line interface (CLI) and a Python application programming interface (API) expose the commands `inspect`, `validate`, `prepare`, `train`, and `predict`; Additional file 2 documents installation, configuration, and outputs. Interfaces for 3D, reaction, mixture, and protein inputs exist but are not evaluated here. The test suite (61 tests) covers configuration parsing, featurisation, feature analysis, the dataset advisor, and training.

Feature evidence and dataset audit

Training-derived scorecard statistics are computed on the training partition. For each feature the scorecard records provenance, including domain, source and definition, training support (number of molecules with a non-zero value),

training mean, standard deviation and range, the Spearman correlation with each target, collinear partners ($|r| \geq 0.95$ by default) and, when supplied, bootstrap evidence for prior coefficients and model-dependent group contributions such as permutation or group-removal effects. An ordered rule assigns each feature an evidence class that can be, *out of distribution/unsupported* (zero support or held-out values outside the training range), *weak/uncertain*, *redundant*, *supported* (a sign-consistent prior coefficient whose bootstrap 95% interval excludes zero, together with a positive model contribution in at least 80% of repeats) or *model-dependent*. Because the out of distribution class uses held-out ranges, it serves post hoc diagnosis only and is never used for selection; it is a simple range based applicability-domain check [29, 30].

The scorecard and recommender serve complementary but distinct purposes. The scorecard is an evidence-documentation and diagnostic layer: it records what is known about a feature under the current dataset and split, but does not itself select features or decide how a feature block enters a model. The recommender is an optional candidate-generation heuristic that returns a compact subset under a fixed relevance and redundancy rule. In particular, the recommender does not consume the scorecard's categorical evidence classes, learn endpoint-specific weights, or automatically alter model architecture. The user remains responsible for deciding whether and how a proposed subset is used.

The dataset advisor (`inspect`) reports SMILES validity, repeated SMILES strings, scaffold diversity [62], Tanimoto diversity of Morgan fingerprints [3, 63], target skewness, outliers, and class balance, and drafts a starting configuration. For this study, an audit script in the repository additionally matched structures by the International Chemical Identifier key (InChIKey) [64]; a duplicated structure was counted as a conflict if its labels disagreed (classification) or spanned more than half the target standard deviation (regression). Covariate shift between partitions was summarised as the number of descriptors with any test value outside the training range and the fraction of test molecules affected. We report both illustrative split-level diagnostics and the mean and population standard deviation over the ten saved scaffold splits where available. For BBBP, the audit also identified duplicate InChIKeys crossing partition boundaries in several scaffold manifests. Inspection of the affected records showed tautomeric representations that produced different Murcko scaffold strings, so scaffold grouping alone did not guarantee structure identity across partitions.

Predictive roles

A route maps named feature blocks to a task, so the same block can be reassigned without changing the feature calculator or training loop. In the *direct* role, features are encoded by an MLP and optionally concatenated with the graph representation; in the *prior* role, a feature set x_p defines a baseline fitted on the training partition; in the *residual* role, a network corrects that baseline:

$$\hat{z}(\mathscr{G}, x_d, x_p) = h_\theta([\,\phi_\theta(\mathscr{G})\,;\,\psi_\theta(x_d)\,]) + f_p(x_p), \tag{1}$$

where ϕ_θ encodes the molecular graph $\mathscr{G}$, ψ_θ encodes directly routed descriptors x_d, h_θ is the readout and output head, and either encoder can be disabled. For regression $\hat{y} = \hat{z}$; for binary classification $\hat{z}$ is a logit and $\hat{y} = \sigma(\hat{z})$. The pattern follows Δ-machine learning [49], with a statistical baseline of physicochemical descriptors in place of a lower-level calculation. The framework fits priors by ridge, least-squares, or logistic regression, optionally with fixed coefficients; in the benchmark, priors were fitted with scikit-learn [65] (ridge, $\alpha = 1$; logistic, $C = 1$, using the log-odds from `decision_function`) on six descriptors (`MolWt`, `MolLogP`, `TPSA`, `NumHDonors`, `NumHAcceptors`, `NumRotatableBonds`), frozen, and supplied as an input. The network was trained on the original target (Huber loss [66] or binary cross-entropy on logits). Listing 1 shows how roles are declared.

Feature recommendation and comparison selectors

The framework recommendation at budget K (`select_evidence_recommendation`) uses the training partition only: (1) compute the absolute Spearman correlation s_j of each feature with the target, setting $s_j = 0$ for features

Listing 1. Configuration excerpt showing the three predictive roles in Eq. (1). The excerpt is validated against the framework schema. Removing the task prior gives a direct hybrid, disabling the descriptor branch gives a prior and residual model, and setting the graph encoder to none gives a descriptor-only model.

```
features:
  physchem: # six descriptors used by the prior
    type: rdkit_descriptors
    names: [MolWt, MolLogP, TPSA, NumHDonors,
            NumHAcceptors, NumRotatableBonds]
  rdkit2d: # full RDKit 2D descriptor pool
    type: rdkit_descriptors
    families: [all]
priors:
  logS_prior: {use: [physchem], fit: ridge} # prior role
routes:
  logS: [rdkit2d] # direct role
tasks:
  logS: {target: logS, kind: scalar, route: logS, prior: logS_prior}
split: {method: scaffold, fractions: [0.8, 0.1, 0.1], seed: 1}
model:
  structure: {encoder: dmpnn, hidden_dim: 128, depth: 3} # graph branch
  descriptors: {enabled: true}
  fusion: concat
  prior: frozen
```

with standard deviation below 10^{-6}; (2) sort features by s_j; (3) add a feature if $s_j \geq 10^{-4}$ and its absolute Pearson correlation with every selected feature is at most 0.85; (4) if fewer than K features pass, pad with the next-ranked features. Padding was never triggered: at least 128 descriptors passed step 3 in every split. The rule is a transparent relevance and redundancy heuristic related to correlation based and minimum redundancy selection [39, 40]; it uses the scorecard's association and collinearity statistics but not its evidence classes. It is therefore presented as a reproducible candidate-generation rule for this workflow, not as a novel feature-selection algorithm or a claim that its selected descriptors are universally important.

Five selectors served as comparators at $K \in \{5, 10, 20, 50\}$: Spearman top-K; mutual information (MI) top-K with the k-nearest-neighbour estimators of scikit-learn ($k = 3$) [37, 38]; an MI-Pearson selector that greedily maximises mutual information relevance while penalising mean absolute Pearson correlation with selected features; Random Forest impurity importance (100 trees, depth ≤ 12) [43]; and L1 selection, where L1 denotes absolute-value regularisation (order of entry on the least angle regression (LARS) lasso path for regression, or absolute coefficients of an L1-penalised logistic regression with $C = 1$ for classification) [41, 42]. L1 selection used a standard scaler fitted on the training partition only. Random subsets (ten draws per split) and the full pool were additional references on ESOL, FreeSolv, and BBBP. Selected subsets were evaluated with a Random Forest (100 trees). Stability was the mean pairwise Jaccard index of the selected sets over the ten splits [46, 47].

Evidence-guided descriptor routing

To test whether feature evidence can guide model configuration, the 210-descriptor input of the direct hybrid was replaced by a compact route. For each split, the recommended subset was evaluated at $K \in \{10, 20, 50\}$, and the budget with the best primary validation metric (K^*) was retained. Spearman, mutual information (MI), and random subsets reused the same split-specific K^* as matched controls; the random route averaged ten random-subset draws per split. An auxiliary prior and residual model also used the recommended K^* subset as its prior input (`prior_rec`); because this changes the predictive role as well as the descriptor route, it is retained in the dataset JSON outputs rather than in the matched direct-hybrid selector table. All compact-route checkpoints were selected by minimum validation loss, and K^* was selected using the primary validation metric evaluated at that checkpoint. The graph-only and full-descriptor references used the same validation-loss checkpoint rule. Thus, checkpoint selection is harmonised across all five datasets and all routing comparisons. The full checkpoint-level audit covers 250 model runs and is

Table 9. MoleculeNet datasets, prediction endpoints, primary metrics, and saved split manifests. Counts refer to the processed datasets used in the experiments.

Dataset	Endpoint	Task	Metric	N	Train	Valid.	Test	Split
ESOL	Aqueous solubility ($\log_{10}$ mol L^{-1})	Regression	RMSE	1,128	902	113	113	Scaffold
FreeSolv	Hydration free energy (kcal mol^{-1})	Regression	RMSE	642	513	64	65	Random
Lipophilicity	Octanol/water $\log D$ at pH 7.4	Regression	RMSE	4,200	3,360	420	420	Scaffold
BBBP	Blood-brain barrier penetration	Binary class.	AUROC	2,039	1,631	204	204	Scaffold
BACE	BACE-1 inhibition	Binary class.	AUROC	1,513	1,209	152	152	Scaffold

recorded in `benchmarks/results/routing_checkpoint_audit.csv`; dataset-specific result files are listed in Additional file 1, Table S13.

Datasets and splits

Five datasets from MoleculeNet were used (Table 9): the ESOL aqueous-solubility benchmark [68], FreeSolv hydration free energy [69], Lipophilicity [70], blood-brain barrier penetration (BBBP) [71], and the beta-site amyloid precursor protein cleaving enzyme 1 (BACE-1) inhibition dataset, referred to as BACE [72], in their DeepChem distributions; SMILES that RDKit could not parse were removed (11 in BBBP), and no deduplication was applied in the primary benchmark. Structures were additionally matched by InChIKey to audit duplicate allocation across the saved partitions. For BBBP, we performed a sensitivity analysis after removing the groups with conflicting labels and retaining one record for each remaining InChIKey. New scaffold split manifests were generated and reused across every model in that sensitivity comparison. Each primary dataset has ten saved 80/10/10 manifests reused by every model. Scaffold splits group molecules by Bemis-Murcko scaffold [62], with acyclic molecules as singleton groups; groups are shuffled with the split seed, sorted by decreasing size, and assigned to the partition whose size is lowest relative to its target fraction. Unlike the DeepChem splitter used for MoleculeNet, large scaffold groups can fall into validation or test, so scores are not directly comparable with published scaffold split values. MoleculeNet recommends random splits for ESOL, FreeSolv, and Lipophilicity [8]; we used scaffold splits for ESOL and Lipophilicity to test generalisation beyond the training scaffolds, which is known to be harder but still optimistic relative to prospective use [19, 21].

The regression targets remain on their native scales. Source-level target summaries, class prevalence, and dataset provenance are provided in Additional file 1, Table S1a. The fixed modelling settings are listed in Additional file 1, Table S1b.

The five benchmark endpoints were modelled independently. Thus, this study evaluates a common multi-representation workflow across several tasks, rather than joint multi-task learning with parameters shared across multiple endpoints.

Reference models and external comparator

Reference models were a dummy predictor, an extended-connectivity fingerprint with radius 2 and 2,048 bits (ECFP4), RDKit 2D descriptors with a Random Forest (100 trees), and a three-layer descriptor MLP. The external comparator was Chemprop v2.3.1 [48] (hidden size 300, depth 3, up to 50 epochs with early stopping on validation loss), as a graph-only D-MPNN and with the 210 descriptors as molecule-level features (x_d). Five framework configurations isolated feature roles: `prior_only`, `direct_only` (descriptor MLP), `residual_only` (graph only), `direct_hybrid` (graph plus 210 descriptors), and `prior_residual` (frozen six-descriptor prior plus graph residual). Feature scalers, target scalers, and priors were fitted on training data only; hyperparameters were fixed a priori (Additional file 1, Table S1b).

Evaluation and statistical analysis

The primary metrics were test root mean squared error (RMSE) for regression and area under the receiver operating characteristic curve (AUROC) for binary classification. Secondary metrics were mean absolute error (MAE), coefficient of determination (R^2), and average precision (AP), reported in Additional file 1. Results are means and population standard deviations (SDs) over ten splits. Because methods share splits, comparisons use paired differences $\Delta_s = m_A(s) - m_B(s)$, summarised by their mean, standard deviation, win count, and a 95% confidence interval (CI) based on a t distribution with nine degrees of freedom. Overlapping training sets make such intervals optimistic [23, 22], so they are read descriptively, without multiplicity correction, and an interval that includes zero is treated as inconclusive rather than as equivalence [25]. Training-seed variance was estimated by repeating four configurations with three seeds on one BBBP split.

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